

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

Guide 1: Swap

University of Minho

2026-2027

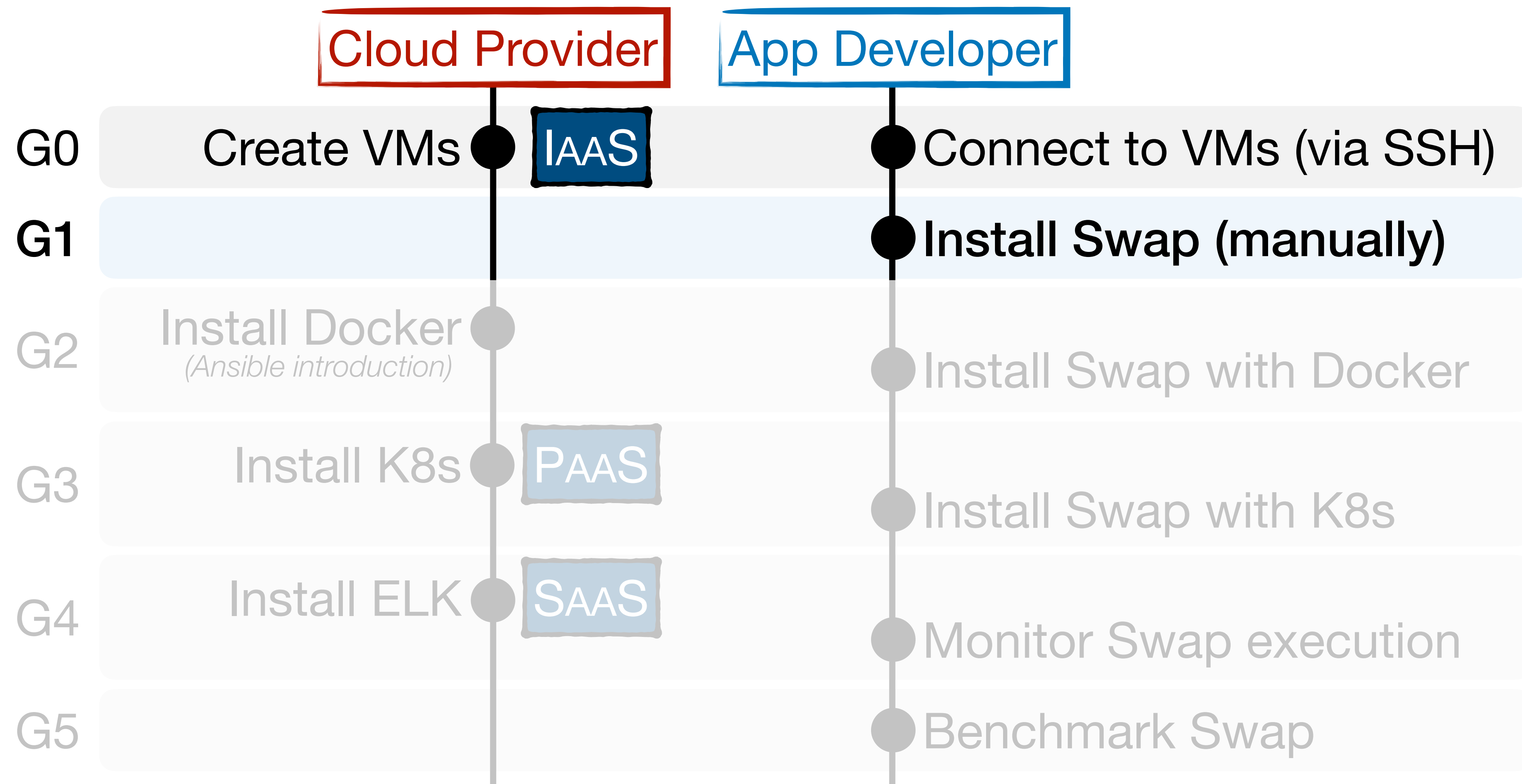


Context

- With “Guide 0”, you (as a *cloud provider*) have set up several VMs for potential cloud users.
- Now, how can an application developer use these VMs to install and deploy applications?
- The goal of this guide (as an *application developer*) is to install and configure a distributed application (**Swap**) on these VMs.

Road Map

Where are you on the roadmap?

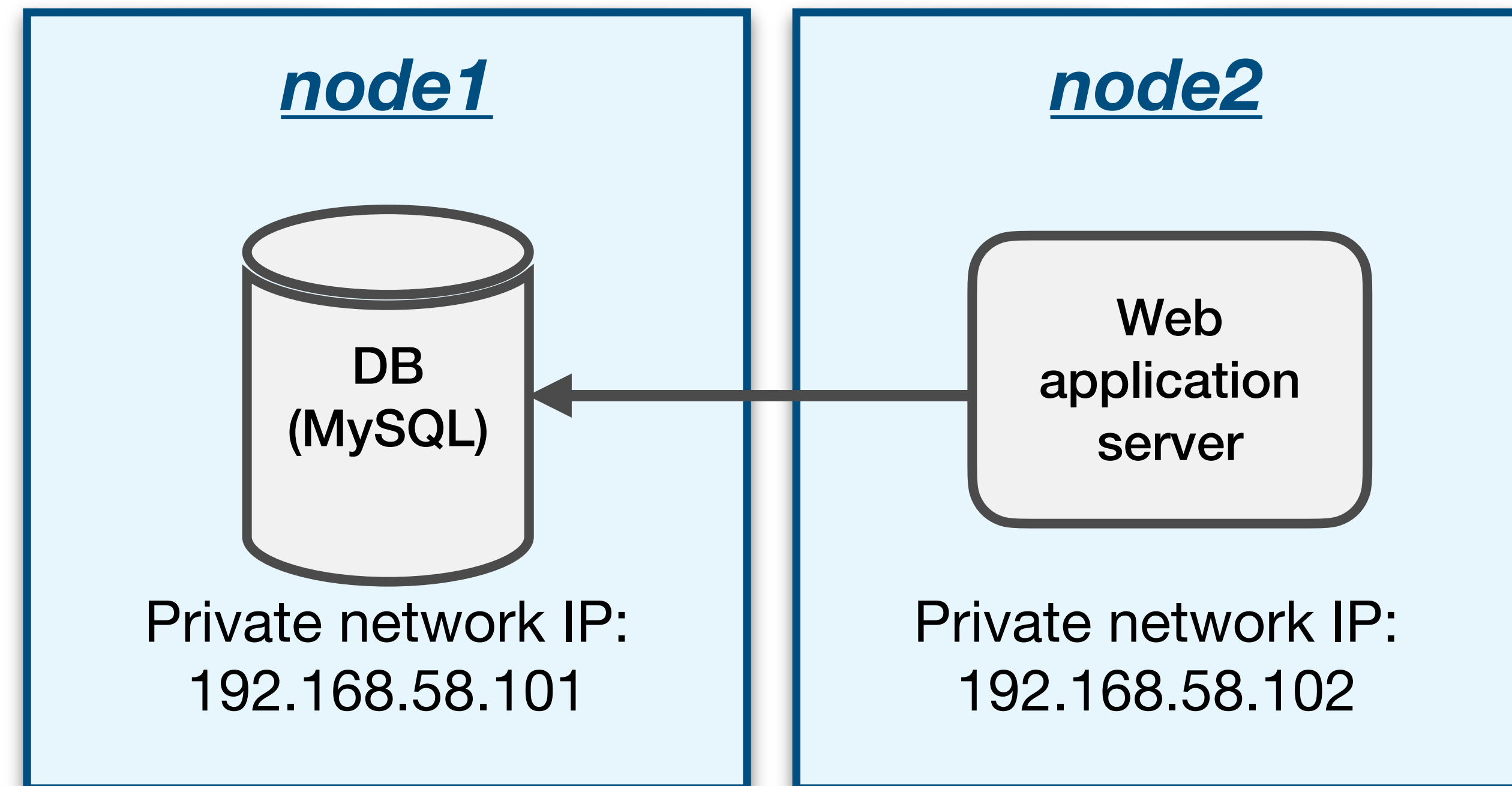


Goal

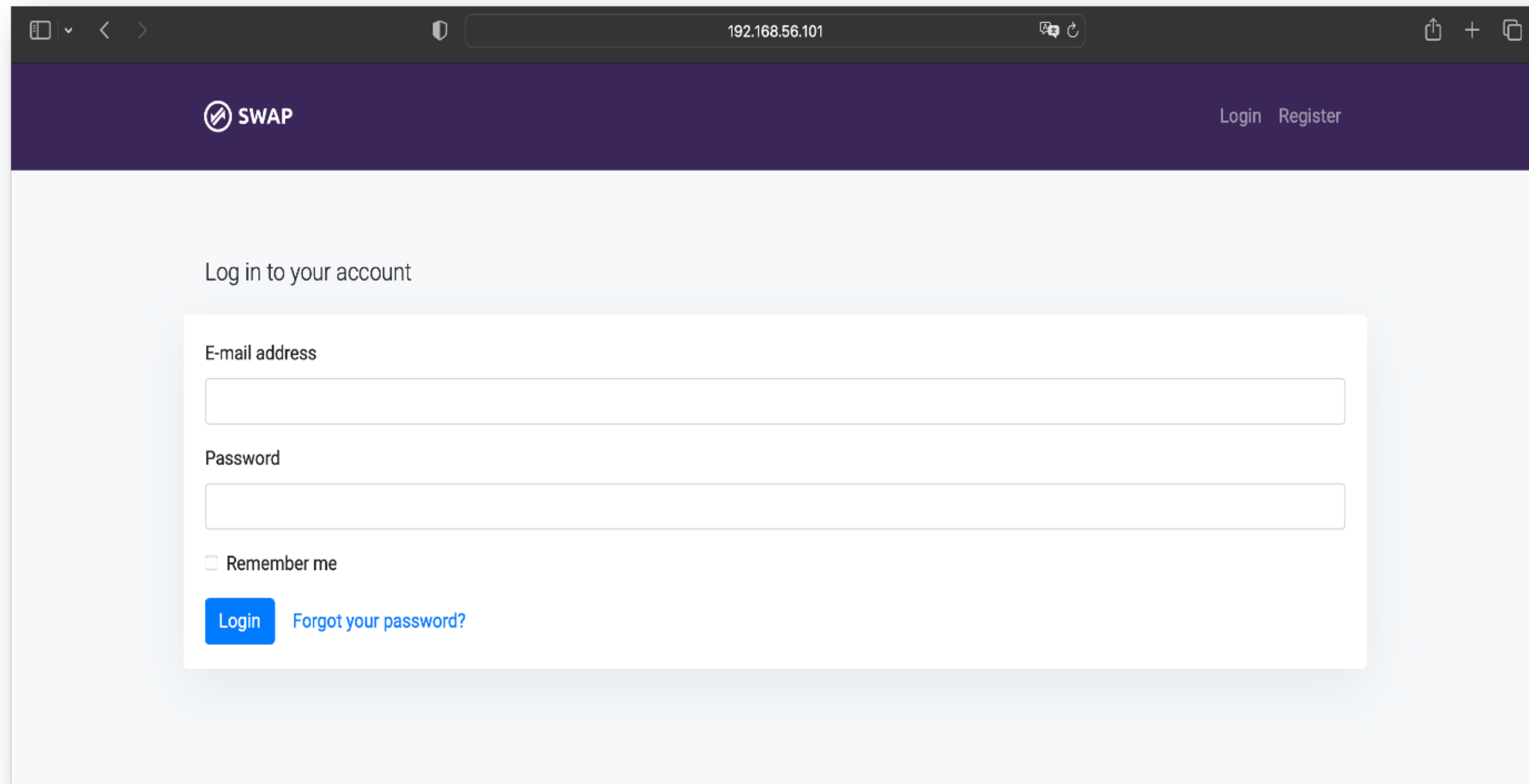
App Developer

● Install the Swap application

- Swap's database will run on *node1*
- Swap's web application server will run on *node2*



Swap: Basic Usage

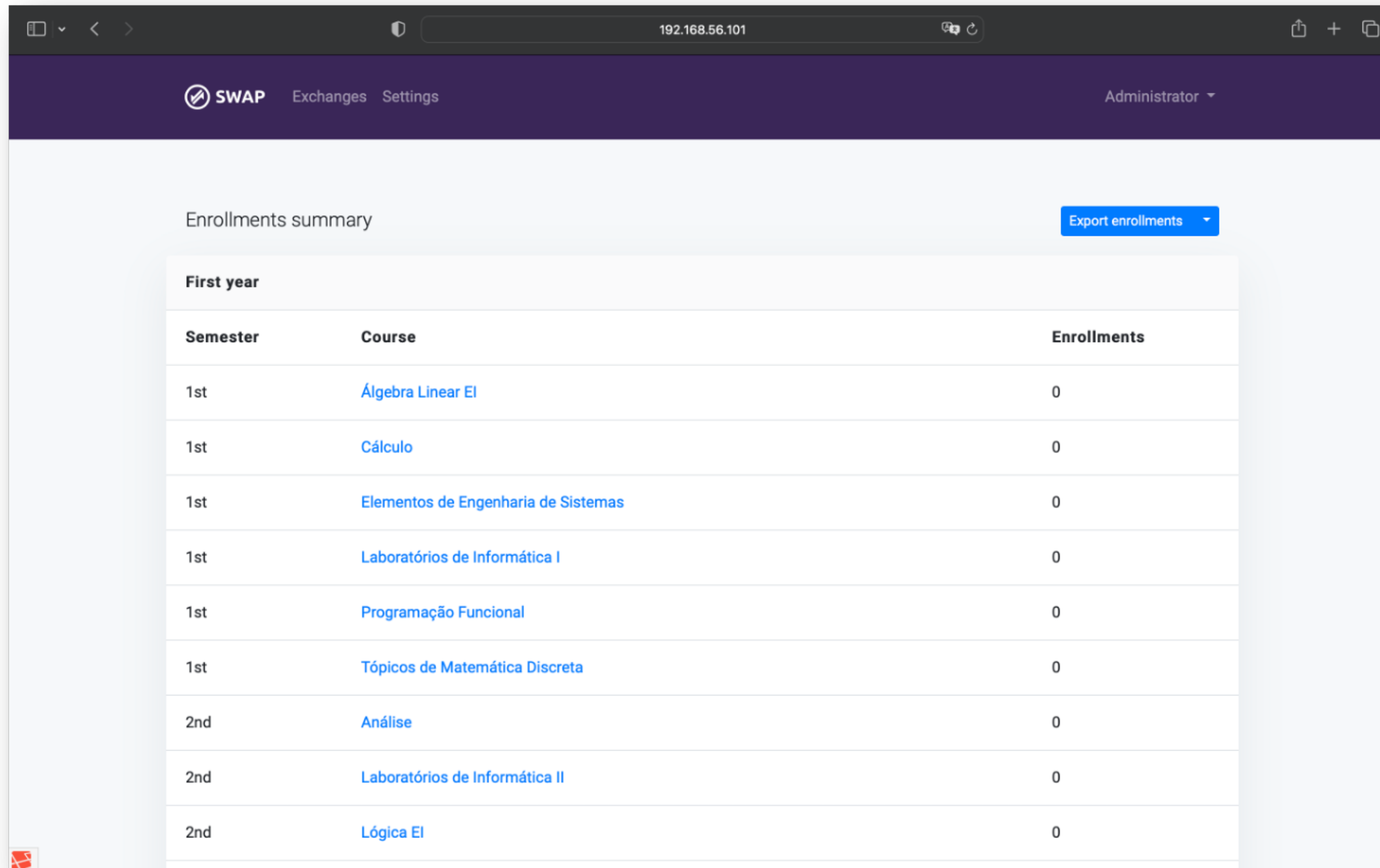


The screenshot shows a web browser window with the address bar displaying '192.168.56.101'. The page has a dark purple header with the 'SWAP' logo on the left and 'Login Register' links on the right. The main content area is light gray and contains the text 'Log in to your account'. Below this is a white login form with two input fields: 'E-mail address' and 'Password'. There is a 'Remember me' checkbox below the password field. At the bottom of the form are a blue 'Login' button and a blue link 'Forgot your password?'.

Email address: contact@hackathonners.org

Password: "123456"

Swap: Basic Usage



The screenshot shows a web browser window with the URL 192.168.56.101. The page has a dark purple header with the SWAP logo, navigation links for 'Exchanges' and 'Settings', and a user role indicator 'Administrator'. Below the header, the main content area is titled 'Enrollments summary' and includes a blue 'Export enrollments' button. A table displays enrollment data for the 'First year', categorized by semester and course. All enrollment counts are currently zero.

Enrollments summary			Export enrollments
First year			
Semester	Course	Enrollments	
1st	Álgebra Linear EI	0	
1st	Cálculo	0	
1st	Elementos de Engenharia de Sistemas	0	
1st	Laboratórios de Informática I	0	
1st	Programação Funcional	0	
1st	Tópicos de Matemática Discreta	0	
2nd	Análise	0	
2nd	Laboratórios de Informática II	0	
2nd	Lógica EI	0	

Swap: Advanced Deployment

